Qualitative Predictors and ANCOVA

Applied Linear Models I • Lecture 11 of 16

Applied Linear Models I

Today

- **Dummy coding** of categorical predictors.
- **ANCOVA**: a continuous covariate + a factor.
- Parallel lines vs **separate slopes** (interaction).
- Testing equal slopes and a single common line.
- Factors with > 2 levels.

Dummy coding

A factor with g levels $\rightarrow g - 1$ indicators; the omitted level is the **baseline**.

`mtcars: mpg ~ am`

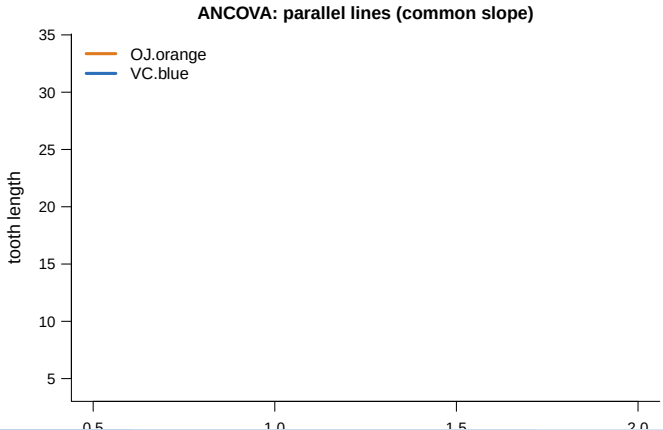
- Intercept = baseline (automatic) mean mpg.
- `am_fmanual` = manual − automatic difference.
- Verified against the raw group means.

Baseline choice is cosmetic: it changes coefficient values, not the model or the factor's F-test.

ANCOVA: parallel-lines model

$$y = \beta_0 + \beta_1 x + \beta_2 d + \varepsilon, \qquad d = \mathbb{1}\{g = \text{level 2}\}.$$

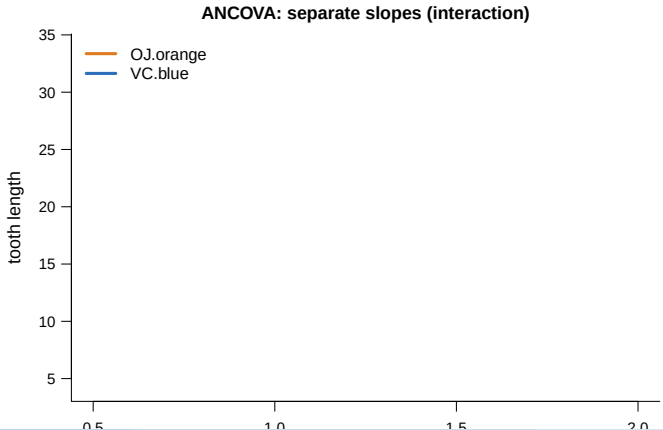
Common slope β_1; the factor shifts the intercept by β_2.



Separate slopes: the interaction

$$y = \beta_0 + \beta_1 x + \beta_2 d + \beta_3 (x \cdot d) + \varepsilon.$$

Group 2 slope $= \beta_1 + \beta_3$, intercept $= \beta_0 + \beta_2$.



Two key tests (ToothGrowth)

Equal slopes? $H_0 : \beta_3 = 0$

`anova(parallel, separate)`: rejects, $p \approx 5 \times 10^{-4}$ – the dose slope differs between OJ and VC.

A single common line? $H_0 : \beta_2 = \beta_3 = 0$

`anova(common, separate)`: rejects – supplement type matters.

When the interaction is significant, there is no single "group effect" – it depends on the covariate value (report the two lines).

Factors with > 2 levels

Three species $\rightarrow$ two dummies; each is the intercept shift vs the baseline (setosa), holding `Petal.Length` fixed.

Test the factor jointly

Use the **joint** F for all dummies (`anova` vs the no-factor model), *not* the individual t's. `iris`: $p \approx 6 \times 10^{-13}$.

A factor-only model (no covariate) is exactly **one-way ANOVA**.

Summary

- Dummies: $g - 1$ indicators, shifts from a baseline; baseline is cosmetic.
- ANCOVA: common slope (parallel) or covariate×factor (separate slopes).
- Test equal slopes (interaction F) and a common line (joint F).
- Multi-level factors need a joint F; factor-only = ANOVA.

Next

Lecture 12: model selection and internal validation (AIC, BIC, C_p, CV).